WAPOR WORLD CONGRESS OF PUBLIC OPINION RESEARCH

# Factorial Survey Experiments

## Design & Analysis — A Practical Workshop

**Francisco Olivos**
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*Based on Auspurg, K. & Hinz, T. (2015). Factorial Survey Experiments. SAGE.*

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A BIT ABOUT ME

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## Deserving a Just Pension: A Factorial Survey Approach*


Juan Carlos Castillo, *Pontificia Universidad Católica de Chile*
Francisco Olivos, *The Chinese University of Hong Kong*
Ariel Azar, *University of Chicago*



*Objective.* This study analyzes which characteristics of pension recipients are taken into account when evaluating the fairness of pensions. Furthermore, it identifies some respondents' characteristics and preferences that could be related to the justice evaluation of different pension amounts. *Methods.* A factorial survey was designed to simultaneously analyze the association of respondents' and recipients' characteristics with the pensions' justice evaluation. *Results.* Findings indicate that although there is a consensual demand for larger pensions, it is still believed that pensions should be allocated primarily based on individual achievement. *Conclusions.* Although in general, larger pensions are on average considered as more just, the justice criteria rely heavily on individual achievement over redistributive considerations, showing willingness to accept very low pensions for those considered not deserving them.



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Research paper

### Teachers' beliefs about educational quality: Examining the power of school settings through survey experiments


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HIGHLIGHTS

- Study aims to characterize teachers' beliefs on determinants of educational quality and the influence of school settings.
- A factorial survey experiment with a probabilistic sample Chilean primary school teachers was used to identify causal attributions of quality.
- Results indicate that teachers believe that decisional power and income satisfaction are the most important determinants of relational quality.
- Planning preparation and a multidisciplinary team of professionals are the most crucial factor in the quality of learning methodology.
- Moreover, school SES, school type, and average student achievement affect teachers' causal beliefs.

## Disentangling Stereotypes towards Older Age Groups: Evidence from Factorial Survey Experiments in China and the USA


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Poetics
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### Untangling causal beliefs: A lay theory of happiness determinants using a factorial survey


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ARTICLE INFO


*Keywords:*
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Causal beliefs
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Chile


ABSTRACT


After two decades of research, the evidence in cross-cultural studies has shown that the lay theories of happiness are not universal. Significant variations have been found in the beliefs about the sources of happiness between countries. However, heterogeneities within countries have been overlooked. In this article, I documented a folk theory of happiness beliefs for the Chilean case. Using factorial surveys, I estimated (a) causal beliefs of happiness, (b) the contingency of the belief about income upon other domains, and (c) the heterogeneity of these beliefs when considering respondents' experience on these dimensions. Health and income are the most prominent determinants of happiness in the belief system of Chileans. For the first time, income is reported as a prominent source of happiness from the perspective of laypeople. Nevertheless, this belief is contingent upon several of the other sources of happiness. In addition, the experience of long-exposure life circumstances explains the heterogeneity of beliefs about lifestyle and partner relationships. Overall, these findings compel us to re-evaluate other once-believed collectivistic countries that have gone through structural changes like Chile. Finally, the potentialities of using factorial surveys for cultural analysis are highlighted.



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Poetics
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### Stratification of educational quality judgments: Insights from two factorial survey experiments on socioeconomic differences in student and parent evaluations


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ARTICLE INFO


*Keywords:*
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ABSTRACT


The use that agents do of cultural knowledge to navigate institutions is a major explanation of inequalities. Nevertheless, the difficulties accessing culture knowledge have led sociologists of education to often rely on declarative forms of culture to gauge explanations on inequalities. Based on the case of Chile, this study contributes to educational inequality research by using factorial survey experiments to assess the stratification of evaluative principles of educational quality or what is termed "judgments of the game." Different teaching and school resources features were experimentally manipulated and presented to representative and probabilistic samples of parents and primary school students to measure their evaluative principles of educational quality. Findings show that socially advantaged parents and students, and from private schools, tend to evaluate educational quality more critically and attribute stronger effects to vignette features, suggesting cultural knowledge stratification. Thus, this study contributes to cultural sociology, sociology of education, and stratification research by testing one of the central theoretical propositions on the explanations of educational inequalities.

3-HOUR ROADMAP

# Today's Workshop at a Glance

Four parts, each building on the last — from theory to a working analysis.

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1

## Foundations

Concepts, vocabulary, and the logic of the factorial survey (FS) method.

SLIDES · NOW

2

## Design the Sample

Build the vignette universe and draw an efficient vignette sample in R.

COLAB

3

## Field It

Export to Excel → generate paper decks in Word; program the survey in Qualtrics.

MANUAL + GUIDE

4

## Analyze

Multilevel models for vignette ratings nested within respondents.

COLAB

# Where the Method Comes From

**An idea from Paul Lazarsfeld** led Peter Rossi to introduce the factorial survey (FS) method in his 1951 dissertation.

- Respondents judged the social status of hypothetical households on a 9-point scale.
- Households varied systematically in husband's/wife's occupation and education.
- Varying these characteristics let Rossi estimate each one's impact on status judgments.
- He could also see how much respondents agreed — or disagreed — about what matters.

ROSSI'S ORIGINAL VIGNETTE (1951)

*"The husband is a **[civil engineer]** with **[a college degree]**. The wife is a **[schoolteacher]** with **[a high school diploma]**."*

Task: "How would you rate the general standing of this household?" (9-point scale)

The bracketed terms are **dimensions** — their bracketed values are **levels**. More on this vocabulary shortly.

CORE DEFINITION

# What Is a Factorial Survey?

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A factorial survey embeds a **multidimensional experimental design** inside a survey: respondents judge **vignettes** — descriptions of hypothetical situations, objects, or persons whose attributes are varied systematically.

*Because the levels are assigned experimentally, differences in ratings can be attributed to the dimensions themselves.*

## ANATOMY OF A VIGNETTE

"This is a **35-year-old** nurse who works **full-time** and earns **$45,000 per year**."

**Dimension: age** → Level: "35-year-old"  **Dimension: work status** → Level: "full-time"  **Dimension: income** → Level: "$45,000"

# Why Not Just Ask Directly?

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Real-world normative judgments are rarely based on a single factor — people weigh several attributes ***against each other***. Single survey items can't isolate which factors actually drive a judgment; vignettes force the same trade-offs respondents make in real evaluations.

1

## Fair Sentencing

Crime circumstances, offender characteristics, and harm to the victim are weighed together to judge a “fair” sentence.

2

## Welfare & Tax Fairness

What counts as a fair benefit level or tax rate depends jointly on need, contribution, and household situation.

3

## Harassment & Abuse

Whether behavior counts as harassment or abuse depends on the combination of context, intent, and severity.

TERMS YOU'LL USE ALL DAY

# Core Vocabulary

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### Vignette

A single description of a hypothetical person, situation, or object shown to a respondent.

### Dimension

An attribute of the vignette that is varied experimentally (e.g., occupation, age, income).

### Level

One specific value a dimension can take (e.g., “nurse” is a level of “occupation”).

### Vignette universe

The full set of all possible level combinations across all dimensions.

### Vignette sample / deck

The subset of vignettes actually drawn from the universe and fielded to respondents.

### Rating / judgment

The respondent's evaluation of a vignette — the outcome variable in the analysis.

POSITIONING THE METHOD

# FS vs. Classic Split-Ballot Experiments

Both are survey experiments, but they differ in scope, purpose, and design complexity.

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## Split-Ballot Experiments

- Typically a single experimental variation
- Often used to refine question wording or answer format
- Narrow methodological focus

## Factorial Survey Experiments

- Multifactorial design — several dimensions varied at once
- Focused on substantive attitudes and behavioral intentions
- Enables estimation of each dimension's independent effect

**Rule of thumb:** If you're varying more than one thing at once and asking about attitudes or decisions, you're likely designing a factorial survey.

# What Can a Factorial Survey Measure?



### Normative Judgments

What counts as fair, appropriate, or deserved — status, sentencing, welfare, harassment.

### Intended Actions

Hiring a job applicant, deciding whether to relocate, or other behavioral intentions.

### Beliefs About Reality

What respondents think is actually true — e.g., perceived risk, perceived cause.

### Feelings & Thoughts

Emotional or cognitive reactions to hypothetical scenarios.

**Example:** Would you employ this applicant? — varying applicant age, gender, work history, and gap length isolates which factors drive hiring decisions (cf. Abraham, Auspurg, & Hinz, 2010).

GROWING ADOPTION

# A Mainstream Method Today

What began as a dissertation technique is now a standard tool in survey research.

**1951**

Rossi's original dissertation applies FS to household status judgments

↑

Increasing implementation in large-scale household surveys

↑

Rising number of FS articles in leading international journals

A QUICK PREVIEW

# How Does FS Relate to Conjoint Analysis?

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Factorial surveys, conjoint analysis, and choice experiments are close cousins — all vary attributes systematically and ask respondents to evaluate the result.

| | Factorial Survey | Conjoint / Choice Experiment |
|---|---|---|
| **Judgment task** | Rate a single vignette (e.g., 0–100 scale) | Choose or rank among several options |
| **Typical focus** | Normative judgments, beliefs, intentions | Preferences, willingness to pay, choice |
| **Presentation** | One profile at a time | Multiple profiles side by side |

*See Auspurg & Hinz (2015, Ch. 6) for more — today we focus on FS design and analysis.*

# The FS Research Process

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**1**

## Design

Choose dimensions & levels; draw an efficient vignette sample.

Ch. 3

**2**

## Implement

Write vignette text; select respondents, scales, and survey mode.

Ch. 4

**3**

## Analyze

Model hierarchical data — robust SEs or multilevel models.

Ch. 5

**4**

## Compare & Extend

Position FS against conjoint/choice methods; assess validity.

Ch. 6

**Today we will walk stages 1, 2, and 3 hands-on** — design in R, fielding in Qualtrics & Word, and analysis in R.

# The Design We Will Build in R

Before opening Colab, we translate the vocabulary from Part 1 into a concrete design: a study of what makes monthly earnings look fair.

**8**

**Dimensions varied**

Gender, age, education, tenure, children, performance, firm's economic situation, earnings

**7,680**

**Vignettes in the universe**

2 × 4 × 2 × 2 × 4 × 2 × 3 × 10 possible level combinations

**−5 to +5**

**Rating scale**

From unfairly low to unfairly high, with 0 as fair

## HOW A VIGNETTE IS BUILT

We combine two sampling strategies: a D-efficient design, built in R, selects a vignette sample that minimizes correlations between dimensions so each one's effect can be estimated precisely; simple random sampling of levels, handled directly in Qualtrics, draws each dimension independently at each response..

*Each vignette describes one synthetic employee; respondents judge whether that person's pay is fair.*

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# Eight Dimensions, Many Levels

Every vignette combines one level from each dimension below — 3,840 unique combinations in total.

| DIMENSION | LEVELS | # |
|---|---|---|
| **Gender** | Male, female | 2 |
| **Age** | 30, 40, 50, 60 years | 4 |
| **Education** | Vocational training, university degree | 2 |
| **Job tenure** | Short vs. long experience in current job | 2 |
| **Number of children** | None, 1, 2, 3 | 4 |
| **Job performance** | Below average, above average | 2 |
| **Firm's economic situation** | High profits, solid, threatened by bankruptcy | 3 |
| **Monthly gross earnings** | 5 steps from €500 to €15,000 | 10 |

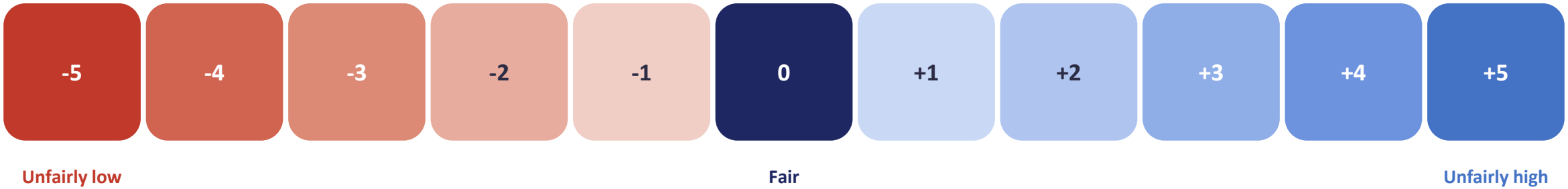
*2 × 4 × 2 × 2 × 4 × 2 × 3 × 5 = 3,840 possible vignettes — the full vignette universe.*

# The Rating Task

Respondents see one full vignette at a time and answer a single fairness question about it.

*"This is a 45-year-old woman with a university degree. She has worked in her current job for a long time, has 2 children, and performs above average. Her firm is highly profitable, and she earns €4,200 gross per month."*

**"Are the monthly gross earnings of this person fair, or are they, from your point of view, unfairly low or unfairly high?"**



NEXT UP

# Designing the Vignette Sample in R

Moving to Google Colab — building the vignette universe, checking D-efficiency, and exporting to Excel.

- Vignette universe & sampling in R
- Excel export → Word decks & Qualtrics
- Multilevel analysis in R